

# Homework 4

## The Coordinate Plane, Lines, and Reading Graphs

Name: \_\_\_\_\_

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### Instructions

This homework is mostly **reading** — it teaches you brand-new material. Read each section **slowly, with a pen**, and answer its questions before moving on. Write your own questions and any “wait, what?” moments in the margin in a different color. You may use the internet or AI as a **teacher** (ask it to explain or to give you extra examples), but always try the questions yourself first.

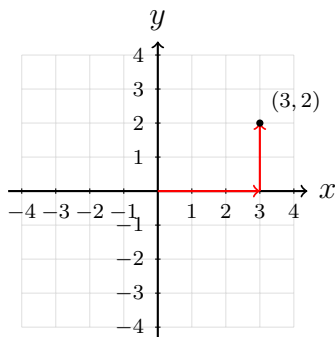
### Reading 1 — The Coordinate Plane

The **coordinate plane** is built from two number lines that cross at a point called the **origin**. The horizontal one is the  **$x$ -axis**; the vertical one is the  **$y$ -axis**.

Every point is named by an **ordered pair**  $(x, y)$ . The first number  $x$  says how far **right** (positive) or **left** (negative) of the origin; the second number  $y$  says how far **up** (positive) or **down** (negative). *Order matters*:  $(2, 5)$  and  $(5, 2)$  are different points.

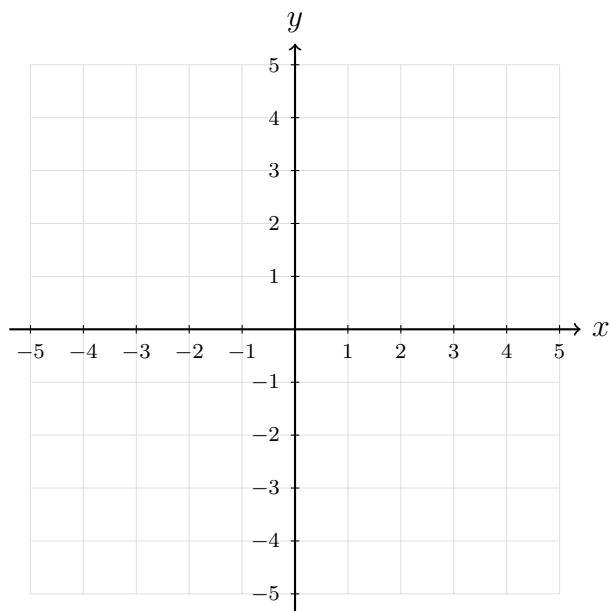
You have seen ordered pairs before: a point  $(x, y)$  is just an **(input, output)** pair — the same kind you used to describe functions. The coordinate plane is simply where we *draw* those pairs.

**Example.** To plot  $(3, 2)$ : start at the origin, go **right** 3, then **up** 2.

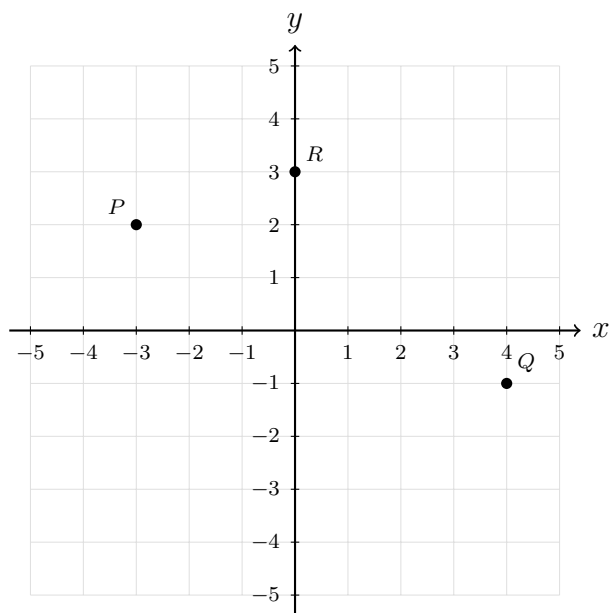


The two axes split the plane into four **quadrants**, numbered I–IV counterclockwise starting from the top-right.

**Q1.** Plot and label these four points on the grid:  $A = (3, 2)$ ,  $B = (-2, 4)$ ,  $C = (0, -3)$ ,  $D = (-4, -1)$ .



**Q2.** Write the coordinates of each labeled point.



$P =$  \_\_\_\_\_  $Q =$  \_\_\_\_\_  $R =$  \_\_\_\_\_

**Q3.** Answer in one sentence each. (a) The point  $(4, 0)$  sits on one of the axes — which one? (b) What is the special name for the point  $(0, 0)$ ? (c) Why are  $(2, 5)$  and  $(5, 2)$  different points?

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## Reading 2 — Graphing a Function

A function takes an input  $x$  and produces one output, written  $y = f(x)$ . To **graph** a function, we turn each input–output pair into a point and plot it:

$$\text{input } x \longrightarrow \text{output } y = f(x) \longrightarrow \text{point } (x, f(x)).$$

The **graph** of a function is the picture of *all* of its  $(x, f(x))$  points. (This is the “graph” representation of a function from our last unit.)

**How to graph one:** (1) pick some inputs; (2) compute their outputs in a table; (3) plot the points; (4) connect them.

**A caution about connecting the dots.** When we connect plotted points with straight segments, we are quietly *assuming* that the function really runs along those segments — that the in-between values lie exactly on our line. For a **linear** function (a straight line) this is true, so connecting the dots is exact. But for any other function it is only an **estimate**: the real graph may curve above or below the line between two points. To see the true shape of a non-linear function, you have to plot *more* points and look closely. (You will meet exactly such a curve in the Challenge.)

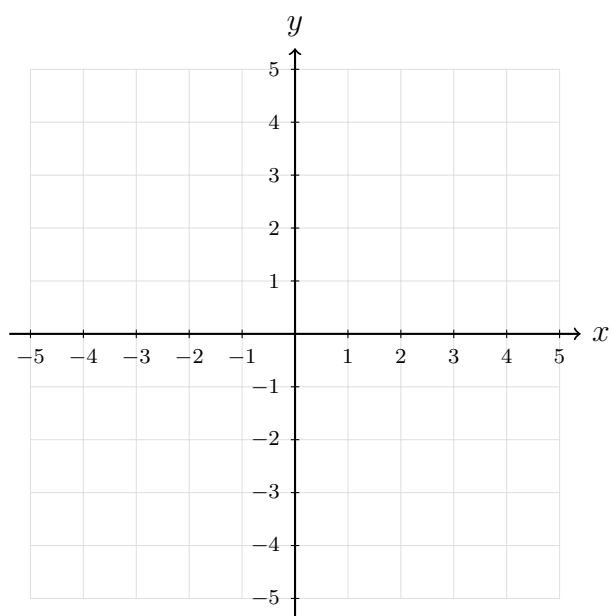
**Example.** For  $f(x) = x + 1$ :

|        |      |      |     |     |
|--------|------|------|-----|-----|
| $x$    | $-2$ | $-1$ | $0$ | $1$ |
| $f(x)$ | $-1$ | $0$  | $1$ | $2$ |

Plotting these points and connecting them gives a straight line. You can try this for yourself.

**Q4.** Let  $f(x) = 2x - 1$ . Complete the table, then plot the points and connect them.

|        |      |      |     |     |
|--------|------|------|-----|-----|
| $x$    | $-2$ | $-1$ | $0$ | $1$ |
| $f(x)$ |      |      |     |     |



**Q5.** Using the line you drew in Q4: (a) what is  $f(3)$ ? (b) for what  $x$  is  $f(x) = 5$ ?

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## Reading 3 — Lines and Slope (Rate of Change)

Some functions graph as perfectly straight **lines** while others don't. The most important number for a line is its **slope** — how steep it is and which way it tilts.

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x}.$$

In words: for every step you take to the *right* (the run), how far *up or down* does the line go (the rise)? From two points  $(x_1, y_1)$  and  $(x_2, y_2)$ ,

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

A **positive** slope goes uphill (left to right); a **negative** slope goes downhill; a **zero** slope is flat. A **vertical** line is special: its run is 0, and you cannot divide by 0, so its slope is **undefined**.

The big idea for later: **slope is a rate of change** — it measures how fast  $y$  changes as  $x$  changes.

**Example.** The line through  $(0, 1)$  and  $(2, 5)$ :  $m = \frac{5 - 1}{2 - 0} = \frac{4}{2} = 2$ . That is, “up 2 for every 1 to the right.”

**Q6.** Find the slope of the line through  $(1, 2)$  and  $(4, 11)$ . Show the rise and the run.

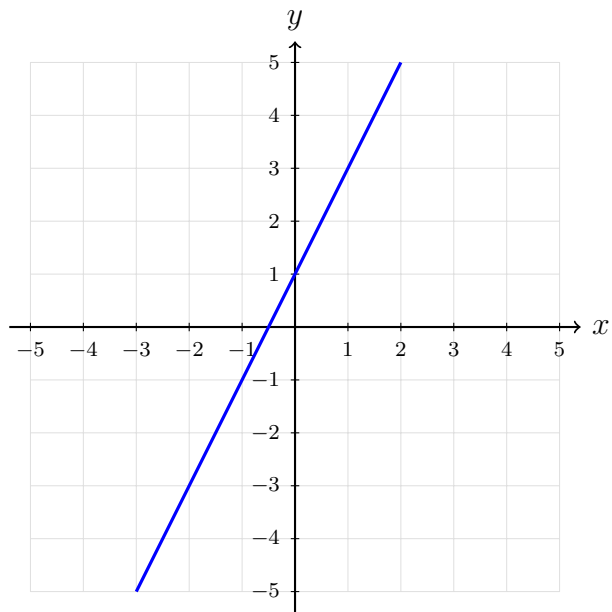
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**Q7.** For each line, is the slope positive, negative, zero, or undefined?

- (a) uphill from left to right \_\_\_\_\_
- (b) flat and horizontal \_\_\_\_\_
- (c) straight up and vertical \_\_\_\_\_
- (d) downhill from left to right \_\_\_\_\_

**Q8.** Find the slope of the line drawn below by reading its rise and run off the grid.



Slope = \_\_\_\_\_

## Reading 4 — What Makes a Function Linear?

We keep using the word **linear** — here is what it actually means.

A function is **linear** when its rate of change (its slope) is the **same everywhere**: no matter which two points you pick, you get the *same* slope. Because the steepness never changes, the graph is a perfectly straight **line**. That single idea can be said three equivalent ways:

- **As a picture:** the graph is a straight line.
- **As an equation:** it can be written  $y = mx + b$  (a constant slope  $m$  and a constant starting value  $b$ ).
- **As a rate:** the rate of change is constant — equal steps in  $x$  always produce equal steps in  $y$ .

**A test you can run on any table.** Step  $x$  up by the same amount each time and watch the outputs. If the outputs change by the *same* amount every step, the function is linear. If those jumps keep changing, it is not.

*Linear* — the steps in  $y$  are constant:

|                 |    |   |   |   |
|-----------------|----|---|---|---|
| $x$             | 0  | 1 | 2 | 3 |
| $f(x) = 2x - 1$ | -1 | 1 | 3 | 5 |

Step changes in  $f(x)$ :  $+2, +2, +2$  — **constant**, so  $f$  is **linear**.

*Not linear* — the steps in  $y$  keep growing:

|              |   |   |   |   |
|--------------|---|---|---|---|
| $x$          | 0 | 1 | 2 | 3 |
| $g(x) = x^2$ | 0 | 1 | 4 | 9 |

Step changes in  $g(x)$ :  $+1, +3, +5$  — **keeps growing**, so  $g$  is **not** linear.

**Non-linear functions.** When the rate of change *keeps changing*, the graph bends — it is a **curve**, not a straight line. (Careful with words: a “line” is *always* straight, so we do not say “non-linear line” — we say **non-linear function**, and its graph is a curve.) The function  $g(x) = x^2$  is the classic example: in the Challenge you will see its average rate of change is 3 between  $x = 1$  and  $x = 2$  but 5 between  $x = 2$  and  $x = 3$ . The slope is different in different places, so the graph curves — and handling that changing rate is exactly what calculus is built for.

**Q.** For each table, find the change in  $y$  at each step, then say whether the function is **linear** or **non-linear** and how you know.

(a)

|     |   |   |    |
|-----|---|---|----|
| $x$ | 0 | 1 | 2  |
| $y$ | 4 | 7 | 10 |

(b)

|     |   |   |   |
|-----|---|---|---|
| $x$ | 0 | 1 | 2 |
| $y$ | 1 | 2 | 5 |

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**Q.** Which of these are linear? Circle them, and for each one that is not, say what makes it non-linear.

$$y = 3x + 2, \quad y = x^2, \quad y = -x, \quad y = \frac{1}{x}.$$

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## Reading 5 — The Equation of a Line: $y = mx + b$

Every non-vertical line can be written in the form

$$y = mx + b,$$

where  $m$  is the **slope** and  $b$  is the  **$y$ -intercept** — the  $y$ -value where the line crosses the  $y$ -axis (the output when  $x = 0$ , the “starting value”).

**To graph from  $y = mx + b$ :** start at the point  $(0, b)$  on the  $y$ -axis, then use the slope to step: right 1, up  $m$  (down if  $m$  is negative). Repeat to get more points, then draw the line.

**Example.**  $y = 3x - 2$ : the  $y$ -intercept is  $b = -2$ , so start at  $(0, -2)$ ; the slope is  $m = 3$ , so go up 3 and right 1.

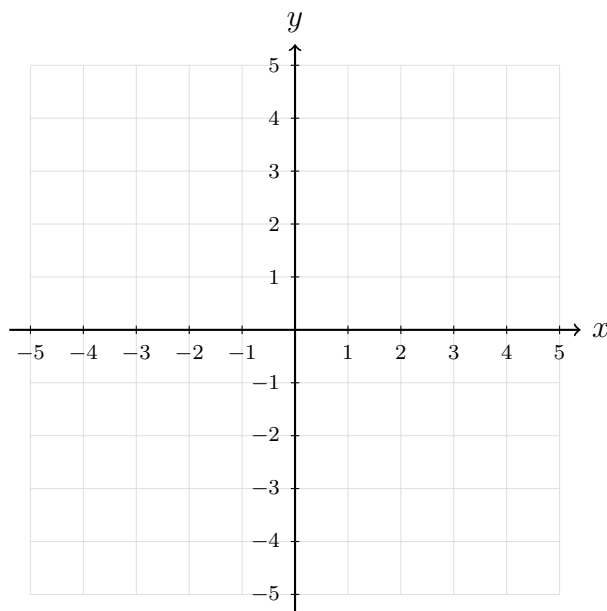
**Reading  $m$  and  $b$  off a graph:** the slope  $m$  is the rise over the run between any two points on the line; the  $y$ -intercept  $b$  is the height where the line crosses the  $y$ -axis.

**Q9.** For  $y = -2x + 4$ :

(a) the slope is \_\_\_\_\_

(b) the  $y$ -intercept is \_\_\_\_\_

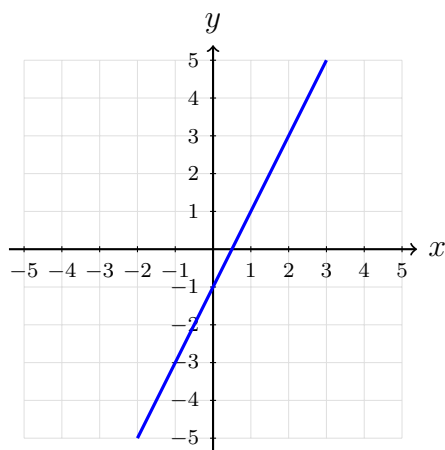
(c) Graph the line on the grid (start at the intercept, then use the slope).



**Q10.** Write the equation  $y = mx + b$  of the line with slope  $\frac{1}{2}$  and  $y$ -intercept  $-3$ .

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**Q11.** Read the slope and the  $y$ -intercept off the line below, then write its equation.



slope = \_\_\_\_\_  $y$ -intercept = \_\_\_\_\_ equation: \_\_\_\_\_

## Reading 5 — Reading Graphs in the Real World

A graph tells a **story**. On a real-world line graph:

- the **slope** is a **rate** — how fast one quantity changes per unit of the other (miles per hour, dollars per item, blocks per minute);
- the  **$y$ -intercept** is a **starting amount** — the value when the other quantity is 0;
- a line going **up** means increasing, **down** means decreasing, and **flat** means not changing.

**Example.** If a graph shows distance (km) against time (hours), the slope is the **speed** (km per hour), and the  $y$ -intercept is how far you had already travelled at time 0.

**Q12. Real-Life Application.** While mining, your total cobblestone  $y$  (in blocks) after  $x$  minutes follows  $y = 5x + 10$ .

(a) What does the slope 5 mean in this situation?

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(b) What does the 10 mean in this situation?

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(c) How much cobblestone do you have after 6 minutes?

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**Q13.** A graph of the water in a tank over time goes **uphill**, then **flat**, then **downhill**. Describe in words what is happening to the water during each part, and what the slope is doing.

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## Challenge — When the Rate Keeps Changing

A straight line has *one* slope everywhere — a single, constant rate. But many graphs are **curves**, and a curve's steepness *changes* from point to point. We can still measure the **average rate of change** between two points, using the very same rise-over-run idea.

Consider  $f(x) = x^2$ :

|        |   |   |   |   |
|--------|---|---|---|---|
| $x$    | 0 | 1 | 2 | 3 |
| $f(x)$ | 0 | 1 | 4 | 9 |

**Q14.** (a) Find the average rate of change of  $f$  between  $x = 1$  and  $x = 2$  (use  $\frac{\text{change in } y}{\text{change in } x}$ ).

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(b) Find the average rate of change between  $x = 2$  and  $x = 3$ .

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(c) Are your two answers the same? What does that tell you about how the *steepness* of this curve is changing as you move to the right?

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*(This question is the doorway to the next unit: when a rate keeps changing, how do we find the rate at a single instant? That is where calculus begins.)*

## Notebook Artifact (Metacognition)

**N1.** On the **left page** of your notebook, write **one Question-Compass** about *slope* (or about graphs), and **one muddiest point** — the single idea from this homework you are least sure about.

**N2.** If you used the internet or AI, describe how you used it.

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